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Ch.2 Bacteria and Viruses

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1: MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Which component is not present in all bacteria?

Answer: (d) Capsule — Ribosomes, cell membrane and nucleoid are basic bacterial structures; a capsule is an extra protective layer found only in some bacteria.

2. What is the typical form of the bacterial chromosome?

Answer: (c) Circular, double-stranded DNA — The main bacterial chromosome is usually a single circular double-stranded DNA molecule located in the nucleoid.

3. Where does respiration occur in bacterial cells?

Answer: (b) Cell membrane — Bacteria lack mitochondria; respiratory enzymes and electron transport are associated with the plasma membrane/mesosomes.

4. Which bacterial group is a good source of antibiotics?

Answer: (d) Actinomycetes — Many actinomycetes, especially Streptomyces, produce antibiotics such as streptomycin and tetracycline.

5. What is the primary function of bacterial flagella?

Answer: (c) Motility — Flagella rotate to move the bacterium through liquid environments.

6. Which bacterial motility is mediated by pili?

Answer: (c) Twitching motility — Type IV pili extend, attach and retract, pulling the cell forward in short jerks.

7. Which structure helps bacteria detect and respond to chemicals?

Answer: (c) Flagella — Chemotaxis is achieved by controlling flagellar rotation in response to chemical gradients.

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8. Which bacterium is not included among the textbook's nitrifying bacteria?

Answer: (d) Pseudomonas — The textbook lists Nitrosomonas, Nitrobacter and Azotobacter with nitrifying bacteria, while Pseudomonas is denitrifying. Note: Azotobacter is commonly described as a nitrogen-fixer in standard biology.

9. Which enzyme converts HIV RNA into DNA?

Answer: (b) Reverse transcriptase — HIV is a retrovirus; reverse transcriptase synthesizes DNA from its RNA genome.

10. What does the HIV capsid contain?

Answer: (b) Single-stranded RNA and reverse transcriptase — HIV carries two copies of single-stranded RNA and the enzyme reverse transcriptase inside its capsid.

SECTION 2: SHORT QUESTIONS

1. Describe the structural components of a bacterial cell wall and their arrangement.

The bacterial cell wall lies outside the plasma membrane and gives shape and protection. Its main framework is **peptidoglycan** (murein). In Gram-positive bacteria the peptidoglycan is thick and contains teichoic acids. In Gram-negative bacteria the peptidoglycan is thin and is covered by an outer membrane containing lipopolysaccharide (LPS), lipoproteins and porins.

2. What is the composition of the peptidoglycan layer?

Peptidoglycan is made of alternating sugar units, **N-acetylglucosamine (NAG)** and **N-acetylmuramic acid (NAM)**, linked in long chains. Short peptide chains attached to NAM cross-link neighbouring chains, forming a strong mesh-like layer.

3. What are mesosomes and what are their functions?

Mesosomes are infoldings of the bacterial plasma membrane. They increase membrane surface area and are associated with respiration, secretion and cell division. In many textbook descriptions they help anchor respiratory enzymes and the bacterial chromosome during division.

4. How can plasmids be used in genetic engineering?

Plasmids are small circular DNA molecules that can replicate independently. Scientists cut a plasmid with restriction enzymes, insert a desired gene, and return it to a bacterium. The plasmid then acts as a **vector**, carrying and expressing the foreign gene, for example to produce insulin or other proteins.

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5. Define sporulation.

Sporulation is the process by which some bacteria form a thick-walled, resistant **endospore** inside the cell when conditions become unfavourable. The endospore can survive heat, drying and chemicals until conditions improve.

6. What is the function of the bacterial capsule?

The capsule is a slimy outer layer, usually polysaccharide. It protects bacteria from drying and phagocytosis, helps them attach to surfaces and host tissues, and can increase virulence.

7. What is the role of pili, and how do they differ from flagella?

Pili	Flagella
Structure: Short, hair-like protein tubes.	**Structure:** Long, whip-like filaments with hook and basal body.
Main role: Attachment, conjugation (sex pili), twitching motility.	**Main role:** Swimming/swarming motility.
Number: Usually many and short.	**Number:** Usually fewer and longer.

8. What are plasmids and how do they help bacteria resist unfavourable conditions?

Plasmids are extra-chromosomal circular DNA molecules. They often carry useful genes such as antibiotic resistance, toxin production or metal resistance. By sharing plasmids through conjugation, bacteria can quickly spread resistance traits in a population.

9. What is the role of endospores in bacterial survival?

Endospores are dormant, highly resistant structures formed by some bacteria. They protect the genetic material during heat, desiccation, radiation and chemical stress. When favourable conditions return, the endospore germinates into a vegetative cell.

10. What is the significance of lipopolysaccharides and lipoproteins in Gram-negative bacteria?

Lipopolysaccharide (LPS) is found in the outer membrane of Gram-negative bacteria. It stabilizes the membrane, acts as an endotoxin and can trigger strong host immune responses. Lipoproteins anchor the outer membrane to the peptidoglycan layer and help maintain cell-envelope structure.

11. How do spirochetes achieve motility?

Spirochetes move using **axial filaments** (endoflagella) located between the cell wall and outer sheath. Rotation of these filaments makes the whole cell bend and rotate, allowing corkscrew-like movement through viscous media.

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12. Differentiate between twitching and gliding movements.

Twitching	Gliding
Mechanism: Extension, attachment and retraction of type IV pili.	**Mechanism:** Smooth movement over surfaces without obvious external appendages; often involves secreted slime/adhesion proteins.
Movement pattern: Short, jerky pulls.	**Movement pattern:** Continuous, smooth sliding.
Surface: Solid surfaces.	**Surface:** Solid or semi-solid surfaces.

13. How do bacteria without flagella achieve motility?

Non-flagellated bacteria may move by twitching with pili, gliding over surfaces, or using axial filaments in spirochetes. Some also spread by swarming differentiation or passive movement, but active surface motility usually involves pili or gliding machinery.

14. Differentiate between swimming and swarming motility.

Swimming	Swarming
Medium: Liquid or very soft agar.	**Medium:** Solid or semi-solid surfaces.
Cells: Usually individual cells with flagella.	**Cells:** Groups of cells, often with many flagella, move together.
Pattern: Directed movement by runs and tumbles.	**Pattern:** Coordinated spreading over a surface.

SECTION 3: LONG QUESTIONS

1. Compare and contrast the cell walls of Gram-positive and Gram-negative bacteria.

Both wall types protect the cell, maintain shape and prevent osmotic lysis, but their envelope architecture differs.

Gram-positive	Gram-negative
Peptidoglycan: Thick, multilayered mesh of NAG–NAM chains cross-linked by peptides.	**Peptidoglycan:** Thin layer in the periplasmic space.
Outer membrane: Absent.	**Outer membrane:** Present; contains lipopolysaccharide (LPS), porins and lipoproteins.
Teichoic acids: Present; give negative charge and help regulate ions.	**Teichoic acids:** Absent.

Gram-positive	Gram-negative
Periplasm: Less prominent.	**Periplasm:** Distinct compartment between plasma membrane and outer membrane.
Gram stain: Retains crystal violet–iodine complex; appears purple.	**Gram stain:** Loses crystal violet during decolorization; takes safranin and appears pink/red.
Sensitivity: Generally more sensitive to penicillin and lysozyme.	**Sensitivity:** Outer membrane acts as an extra barrier to many antibiotics, dyes and detergents.

Thus Gram-positive walls are mainly thick peptidoglycan, whereas Gram-negative walls are a thin peptidoglycan layer plus an additional outer membrane.

2. Explain the different methods of movement in bacteria.

Method	Mechanism	Example/notes
Swimming	Rotation of flagella in liquid; alternating runs and tumbles.	Escherichia coli.
Swarming	Coordinated movement of many flagellated cells over a surface.	Proteus species.
Twitching	Type IV pili extend, attach and retract, pulling the cell in jerks.	Pseudomonas, Neisseria.
Gliding	Smooth surface movement without flagella; involves adhesion proteins and/or slime.	Myxobacteria, cyanobacteria.
Axial filament	Endoflagella rotate within the periplasmic space of spirochetes, producing corkscrew motion.	Treponema, Borrelia.

Flagellar movement is biased by chemotaxis: bacteria lengthen runs toward attractants and increase tumbles away from repellents, so they move up favourable chemical gradients.

3. Explain the structure of a bacterial flagellum.

- **Filament:** long, hollow, helical tube made of flagellin protein; acts as the propeller.
- **Hook:** short, curved, flexible connector that transmits torque from the motor to the filament.

- **Basal body:** motor embedded in the cell envelope; contains a rod and rings. Gram-negative bacteria have L, P, MS and C rings; Gram-positive bacteria have fewer rings because they lack an outer membrane.
- **Motor:** driven by proton motive force (or Na^+ in some species); rotation direction controls runs and tumbles.

The flagellum grows by adding flagellin subunits at the tip, not at the base. Its rotation converts electrochemical energy into mechanical movement.

4. Describe the formation of an endospore in bacteria.

> Unfavourable conditions → DNA replication and asymmetric septum → Forespore engulfment → Cortex and spore coats deposited → Maturation and dehydration → Mother cell lysis releases endospore → Germination when conditions improve

- The core contains DNA, ribosomes, dipicolinic acid complexed with calcium, and small acid-soluble proteins that protect DNA.
- The cortex is a thick peptidoglycan layer; the spore coat is proteinaceous and resistant to chemicals and enzymes.
- Dehydration and mineralization make the endospore highly resistant to heat, drying, radiation and disinfectants.

Endospore formation is a survival process, not reproduction: one vegetative cell produces one endospore, which germinates into one vegetative cell.

5. Briefly describe the ecological and economic importance of bacteria.

Area	Importance
Decomposition	Break down dead organic matter and recycle carbon, nitrogen and minerals.
Nitrogen cycle	Nitrogen fixation, nitrification and denitrification maintain soil fertility.
Food production	Yogurt, cheese, vinegar, pickles and other fermentation products.
Medicine	Antibiotics, vaccines, insulin and other recombinant proteins.
Industry	Enzymes, organic acids, solvents, bioplastics and biofuels.
Environment	Bioremediation, sewage treatment and degradation of pollutants.
Disease	Some bacteria cause plant, animal and human diseases, so control measures are also economically important.

Overall, bacteria are essential for nutrient cycling and biotechnology, although a minority are pathogens.

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6. Explain the use of bacteria in research and technology.

- **Model organisms:** E. coli is used to study genetics, metabolism, gene regulation and DNA replication because it grows quickly and is easy to manipulate.
- **Genetic engineering:** plasmids, restriction enzymes and ligases allow cloning and expression of foreign genes.
- **Protein production:** engineered bacteria produce insulin, growth hormone, vaccines, enzymes and diagnostic proteins.
- **Antibiotic discovery:** actinomycetes such as Streptomyces are screened for antimicrobial compounds.
- **Bioremediation:** bacteria degrade oil spills, pesticides, plastics and industrial wastes.
- **Molecular tools:** bacterial enzymes such as Taq polymerase, restriction enzymes and CRISPR-Cas systems are central to PCR, cloning and genome editing.

7. Define normal flora and state the benefits we get from normal bacterial flora.

Normal flora (normal microbiota) are the microorganisms that normally live on and inside the human body without causing disease under normal conditions. They colonize skin, mouth, intestine and other surfaces.

- They occupy niches and compete with pathogens, preventing colonization (microbial antagonism).
- Gut bacteria help digest food and produce vitamins such as vitamin K and some B vitamins.
- They stimulate and train the immune system.
- They maintain healthy pH and chemical conditions in body habitats.

If normal flora is disturbed, for example by antibiotics, opportunistic pathogens may overgrow.

8. Explain the structure of a model bacteriophage and HIV.

Model bacteriophage (e.g., T4)

- Icosahedral head containing double-stranded DNA.
- Contractile tail sheath surrounding a tail tube; base plate and tail fibres recognize bacterial receptors.
- During infection, the sheath contracts and DNA is injected through the tail tube into the host cell.

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HIV

- Enveloped virus with glycoprotein spikes (gp120/gp41) for attachment to host CD4 receptors.
- Capsid contains two copies of single-stranded RNA and reverse transcriptase.
- Matrix protein lies between the envelope and capsid.
- Reverse transcriptase makes DNA from RNA; integrase inserts viral DNA into the host genome.

Thus bacteriophages are complex DNA viruses of bacteria, while HIV is an enveloped RNA retrovirus of human immune cells.

INQUISITIVE QUESTIONS

1. Why do bacteria have ribosomes even though they lack membrane-bound organelles?

Ribosomes are not membrane-bound; they are ribonucleoprotein particles needed for protein synthesis. All living cells must make proteins, so bacteria contain 70S ribosomes even though they lack organelles such as mitochondria or ER.

2. If bacteria lack mitochondria, how do they generate energy?

Bacteria carry out glycolysis in the cytoplasm and use the plasma membrane for electron transport and ATP synthesis. Respiratory or photosynthetic enzymes are located in the membrane or its infoldings, so a proton gradient can still drive ATP synthase.

3. Why do some bacteria use twitching motility with pili instead of flagella?

Twitching is useful on solid surfaces and in biofilms where flagellar swimming is ineffective. Pili allow strong attachment and pulling, helping bacteria colonize surfaces, move toward nutrients and form communities.

4. Give reasons in favour of “Prevention is better than cure”.

Prevention avoids disease before damage occurs. Vaccination, hygiene, safe water, safe food, vector control and safe behaviour reduce transmission, lower treatment costs and prevent complications. For diseases such as AIDS, once infection occurs it is difficult to eliminate, so prevention is especially important.

5. Correlate social and cultural values with the prevalence of AIDS.

AIDS spread is influenced by awareness, education, stigma, gender norms, sexual behaviour, drug use and access to healthcare. Societies with open health education, testing, condom availability and safe blood practices usually reduce transmission. Stigma and misinformation can hide infection and delay treatment, increasing prevalence.

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